

Black-Box Embedding Inversion Attack on Vector Databases

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Abstract

Vector databases that index and serve dense embeddings have become central to modern data science applications. Embeddings are often regarded as privacy-preserving surrogates for raw data, motivating practitioners to outsource vector databases to third-party services for scalability. However, recent studies show that even text embeddings alone can leak sensitive information, raising serious privacy concerns. Existing attacks on image embeddings, meanwhile, typically assume access to model architecture or parameters, which does not hold in outsourced settings. In this paper, we propose a novel black-box image embedding inversion attack that reconstructs high-fidelity images using only query access to the embedding model or API. Our approach leverages an in-distribution auxiliary dataset to train a conditional diffusion model, capturing domain-aligned knowledge of the data owner's private images. We introduce an embedding-guided cross-attention mechanism, where image embeddings serve as conditional signals to steer the generation process. To improve efficiency, we perform diffusion in the latent space of a pretrained VQGAN with deterministic decoding, which reduces the computational cost while preserving both structural and perceptual fidelity in reconstructed images. Extensive experiments on three real-world datasets and three popular embedding models demonstrate that our approach significantly outperforms four state-of-the-art baselines. These findings reveal that image embeddings can expose sensitive visual information, highlighting the need for stronger privacy protections in outsourced vector databases.

CCS Concepts

• **Security and privacy → Domain-specific security and privacy architectures**;

Keywords

Vector Databases, Embedding Inversion, Privacy Leakage, Black-box Attack, Diffusion Model

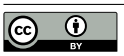
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Resource Availability:
The source code and artifacts of this paper are publicly available at https://doi.org/10.5281/zenodo.20493950. The development repository is available at https://github.com/RUCPrivacy/libra.

1 Introduction

Vector databases have rapidly emerged as key infrastructure for modern data science applications. By indexing high-dimensional embeddings, they enable efficient similarity search for a wide range of downstream tasks, including semantic retrieval [16], recommendation [31, 49], and retrieval-augmented generation [20] for large language models. Commercial systems like Milvus [42] and Faiss [15] offer cloud-managed vector database services that scale

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seamlessly to billions of embedding vectors. In these systems, data owners typically encode their private content, such as text, images, or audio, into embeddings using local models and then outsource these vectors to cloud services for indexing and query, as illustrated in Figure 1. Since the raw data never leaves the data owner's side, this deployment paradigm is often regarded as privacy-preserving and has been widely adopted in practice.

Yet, as vector databases become deeply integrated into online systems, a fundamental question arises: *can embeddings truly protect the privacy of the raw data?* This assumption, while convenient, is increasingly challenged by empirical findings [25, 44]. Recent studies have revealed that dense embeddings can preserve rich information about their original inputs, allowing adversaries to infer or even reconstruct sensitive data. This risk is particularly concerning in the outsourcing paradigm, where the database service provider—or any entity with access to the stored embeddings—might recover private content without needing access to the original data or models.

For instance, prior research [21, 26, 35] has demonstrated inversion attacks on text embeddings by formulating the inversion as a generation task, showing that text embeddings retain substantial lexical and semantic information of the original documents, which can be effectively reconstructed from the embeddings alone. For image embeddings, existing works [11, 19, 22, 34, 44, 50, 51] primarily focus on reconstructing private images from gradients exchanged during the split learning or collaborative inference process, revealing that image embeddings are far from opaque, that is, they encode compressed but recoverable information about their original inputs.

Nevertheless, these image inversion attacks typically rely on a white-box assumption, in which adversaries have access to the model architecture or parameters [2, 9, 27, 48]. This allows the adversary to use gradients and intermediate feature representations as explicit inversion targets, adjusting candidate inputs until their internal signals align with the observed ones to reconstruct the original data. However, such information is unavailable in outsourced settings. The adversary can observe only the final-layer embedding vector and access the embedding model in a black-box manner [26, 38, 48]. Moreover, prior studies have shown limited effectiveness when attacking deeper intermediate representations [11]. For example, when images are encoded through more than seven layers, the reconstruction quality deteriorates drastically [51]. In practice, data owners in outsourced scenarios often employ deep embedding models such as CLIP [30] or ResNet [10], which contain tens or even hundreds of layers to generate image embeddings, rendering previous attacks inapplicable in this setting. Whether meaningful visual reconstruction remains possible under these constraints remains an open and practically important question.

In this paper, we propose a novel black-box image embedding inversion attack framework, namely LIBRA, which reconstructs private images from outsourced embeddings without access to model architecture or parameters. We thus operate under a strict black-box setting. Our key idea is to exploit the remarkable generative capability of diffusion models to synthesize images that align with a given image embedding. However, final-layer embeddings are highly compressed representations that retain only high-level semantics while discarding essential pixel-level information. As a result, the mapping from embeddings to images is an underdetermined problem. Our goal is not pixel-level recovery, but to reveal semantic privacy leakage by reconstructing plausible images with consistent semantics and coarse structure. Consequently, directly adopting techniques like diffusion models to reconstruct images from such embeddings leads to structural drift and texture inconsistencies, undermining the visual fidelity.

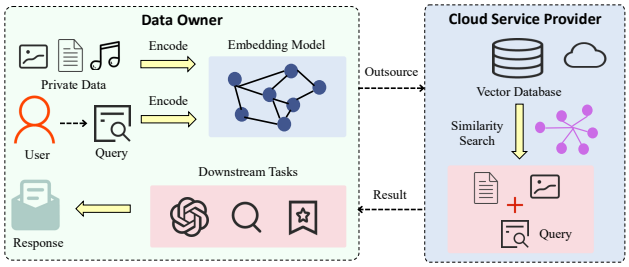


Figure 1: An illustration of outsourcing vector databases.

To address these problems, LIBRA employs an in-distribution auxiliary dataset that is drawn from the same or a similar distribution as the data owner's private images, and trains a conditional diffusion model (CDM) with a novel embedding-guided cross-attention mechanism. Specifically, we first generate image embeddings for the auxiliary dataset utilizing the same embedding model that is used by the data owner. Importantly, LIBRA only requires black-box access to this model, without any knowledge of its internal architecture or parameters. We then train the CDM through iterative forward diffusion and reverse prediction steps. During training, the embeddings are injected as supervision signals at every diffusion timestep to guide the U-Net's cross-attention layers, enabling the model to align with the statistical characteristics of the target embeddings and effectively mitigate structural drift during reconstruction. To further improve training efficiency, LIBRA projects input images into a latent space using a pretrained VQGAN model and performs the diffusion process within this compact representation. This design not only reduces computational overhead but also preserves the structural coherence and perceptual fidelity of the reconstructed images.

We conduct extensive experiments on three real-world datasets using three representative embedding models, and compare our method against four state-of-the-art inversion attacks. Experimental results demonstrate that LIBRA consistently outperforms existing approaches in both quantitative similarity metrics and qualitative visual fidelity. Furthermore, LIBRA exhibits significantly lower inference times, enabling adversaries to launch effective image embedding inversion attacks within a short timeframe. Finally, we discuss several potential defense strategies to mitigate the privacy risks posed by such attacks.

In summary, we make the following contributions.

- We propose LIBRA, a novel black-box image embedding inversion attack framework that reconstructs private images from outsourced final-layer embeddings, without access to model parameters or architecture.
- We present an embedding-guided cross attention mechanism that injects embedding signals at every timestep to train a conditional diffusion model, effectively reducing structural drift and texture mismatch.

- We leverage a pretrained VQGAN model to compress input images into a compact latent space for performing the diffusion process, improving computational efficiency while preserving structural coherence.
- We conduct extensive experiments on three real-world datasets and three embedding models. The experimental results demonstrate that LIBRA surpasses four state-of-the-art baselines in reconstruction fidelity and runtime. We further discuss potential defenses against the attack.

2 Related Works and Preliminaries

2.1 Embedding Inversion Attacks

Existing image embedding inversion attacks (EIAs) can be categorized into white-box and black-box settings, depending on how much access the adversary has to the target model.

White-box EIAs. White-box EIAs assume that the adversary has access to model parameters. Early methods, such as rMLE [11] and LM [34], formulate reconstruction as a likelihood maximization problem, utilizing analytical priors such as total variation (TV) [33] or deep image priors [39] to align candidate inputs with intermediate representations. Recent frameworks also integrate generative priors to enhance fidelity. For instance, GLASS [22] optimizes latent codes within the StyleGAN2 [17] generator. Similarly, DRAG [19] leverages a pre-trained latent diffusion model (LDM) as a prior, iteratively optimizing the candidate image via gradient descent to achieve feature-space alignment with the target embedding. However, their reliance on internal model access and engineered priors renders them ineffective in black-box settings.

Black-box EIAs. Black-box EIAs assume that adversaries can only query a target model without direct access to its parameters or gradients. Early work [11] reconstructs private images by training inverse networks on input–output pairs. Subsequent research has evolved toward more sophisticated generative strategies. [46] introduces a GAN-based reconstruction attack in split learning, aligning shadow and observed feature spaces via adversarial objectives, and further introduces distillation-based client-side attacks to infer data distributions. [6] incorporates feature-map-aware conditioning into a U-Net denoiser to guide black-box reconstruction, while [48] combines pretrained latent diffusion models with optimization-based inversion for improved visual fidelity. Nonetheless, existing black-box EIAs primarily target shallow intermediate representations. Their effectiveness drops significantly as the number of layers in the encoder increases.

2.2 Preliminaries

Diffusion models (DMs) [12, 24, 36] synthesize data by reversing a gradual Gaussian noising process [28], offering superior training stability and mode coverage compared to generative adversarial networks (GANs) and variational autoencoders (VAEs) [7, 32]. Among diffusion-based generative models, denoising diffusion probabilistic models (DDPMs) [12, 28] instantiate this framework as a discrete-time Markovian noising–denoising process, offering stable training and high-fidelity samples. Specifically, the DDPM training procedure consists of two phases: a forward diffusion phase and a backward denoising phase.

Forward Diffusion Phase. Given an original sample $x_0 \sim q(x_0)$, the data is progressively corrupted over T timesteps according to a predefined variance schedule $\{\beta_t\}_{t=1}^{T}$, where $\beta_t \in (0,1)$. The marginal distribution of a noisy sample x_t conditioned on x_0 is expressed as $q(x_t \mid x_0) = \mathcal{N}(x_t; \sqrt{\bar{\alpha}_t}x_0, (1-\bar{\alpha}_t)I)$, where $\alpha_t = 1-\beta_t$ and $\bar{\alpha}_t = \prod_{i=1}^{t} \alpha_i$. Utilizing the reparameterization trick, sampling x_t can be performed in a single step:

$$x_t = \sqrt{\bar{\alpha}_t}\, x_0 + \sqrt{1-\bar{\alpha}_t}\, \xi, \qquad \xi \sim \mathcal{N}(0, I). \tag{1}$$

Eqn (1) allows the model to circumvent iterative simulation during training, enabling efficient supervision of the noise prediction network at arbitrary timesteps.

Backward Denoising Phase. To recover the original data distribution, DDPMs learn a reverse denoising process that approximates the true reverse transition $q(x_{t-1} \mid x_t)$, which is generally intractable due to the unknown data distribution. A neural network is trained to parameterize the reverse conditional distribution, as formulated in Eqn (2):

$$p_\theta(x_{t-1} \mid x_t) = \mathcal{N}\big(x_{t-1};\, \mu_\theta(x_t, t),\, \Sigma_\theta(x_t, t)\big). \tag{2}$$

Here, the mean μ_θ and covariance Σ_θ are parameterized as functions of the noisy input x_t and the timestep t. During inference, starting from pure Gaussian noise $x_T \sim \mathcal{N}(0, I)$, the model iteratively applies the learned reverse transitions to progressively denoise the sample, ultimately yielding high-quality samples that approximate the original data distribution.

3 Problem Statement

3.1 System Model

There are two parties involved in the outsourced vector database setting: the data owner who encodes private images into embeddings, and the cloud service provider that manages these embeddings within the vector database.

Data Owner. The data owner has a private dataset $\mathbf{D}_{\text{pri}} = \{x_i\}_{i=1}^{m} \subset \mathcal{X}$ consisting of a set of m private images. The data owner runs an embedding model $\mathbf{E} : \mathcal{X} \to \mathbb{R}^d$ to compute the embeddings for each image $x \in \mathbf{D}_{\text{pri}}$, where d is the dimension of the embedding, i.e.,

$$e = \mathbf{E}(x), \quad x \in \mathbf{D}_{\text{pri}}.$$

The resulting vectors and metadata are sent to the cloud.

Cloud Service Provider. The cloud service provider manages the vector database on behalf of the data owner. It handles storage, indexing, and query execution over the outsourced embeddings, but does not have access to the underlying raw data in $\mathbf{D}_{\text{pri}}$. For instance, given a query embedding, the cloud can retrieve approximate nearest neighbors using a prescribed similarity metric, such as cosine similarity [5] or Euclidean distance [3], to support downstream tasks such as retrieval-augmented generation.

3.2 Threat Model

We consider a semi-honest threat model, in which the adversary attempts to infer private images from the embeddings, but does not tamper with the embedding model, interfere with the data owner, or inject crafted inputs. In the outsourced setting, the cloud service provider or any party with access to the stored embeddings may act as an adversary [21, 26, 35]. We focus on a black-box

setting in which the adversary has only query access to the data owner's embedding model. The adversary may submit images and observe their corresponding embeddings, but has no knowledge of the model's parameters, architecture, or internal computations. Moreover, we assume that the adversary has access to a set of auxiliary dataset $\mathbf{D}_{\text{aux}} = \{x'_j\}_{j=1}^{n}$ drawn i.i.d. from the same or a similar underlying distribution as the private images, which is a commonly used assumption in inversion attacks [2, 44, 45].

3.3 Problem Formulation

Given a target embedding e, the goal of the adversary is to reconstruct a plausible approximation of its original input x. Formally, the adversary aims to learn an embedding-inversion mapping $\mathcal{F} : \mathbb{R}^d \to \mathcal{X}$ that reconstructs x from e, as defined in Eqn (3)

$$\hat{x} = \mathcal{F}(e), \tag{3}$$

where $\hat{x}$ is the reconstructed approximation of the original image x. The expected reconstruction risk is then minimized as formulated in Eqn (4)

$$\begin{aligned} \min_{\mathcal{F}} \quad & \mathbb{E}_{(x,e)} \left[\mathcal{L}_{\text{rec}}(\mathcal{F}(e), x) \right] \\ \text{s.t.} \quad & e = \mathbf{E}(x), \end{aligned} \tag{4}$$

where $\mathcal{L}_{\text{rec}}(\cdot, \cdot)$ is the reconstruction loss between the original image and the reconstructed image.

4 LIBRA Framework

In this section, we present our attack framework LIBRA. We first give an overview of the framework, then detail the training stage and recovering stage in LIBRA.

4.1 Overview

Fundamentally, the inversion attack is feasible because the final-layer embedding, though highly compressed, preserves high-level semantic information and coarse structural cues of the original image. This means that, despite discarding pixel-level details, the embedding contains sufficient information to constrain plausible reconstructions. Motivated by this, we adopt the conditional diffusion model (CDM) for the attack, leveraging its strong generative capacity and multi-scale refinement ability to progressively synthesize detailed images. However, directly applying CDM on the embedding is insufficient. Without explicit guidance, the generation often drifts structurally and produces texture inconsistencies due to the underdetermined nature of recovering images from highly compressed embeddings. To overcome this, our key idea is to utilize the image embedding as a guiding signal and inject it into the diffusion process. This strategy aligns the generative process with the semantic and structural properties of the target, ensuring reconstructions that are both visually coherent and faithful.

Figure 2 illustrates an overview of our attack framework LIBRA, which consists of two stages: a training stage and a recovering stage. In the training stage, the adversary trains a generator G_θ on the auxiliary dataset $\mathbf{D}_{\text{aux}}$ in the latent space $\mathcal{Z}$ by optimizing a denoising objective, such that the model approximates the conditional distribution of latent representations given embeddings. For each image $x' \in \mathbf{D}_{\text{aux}}$, the adversary computes its corresponding

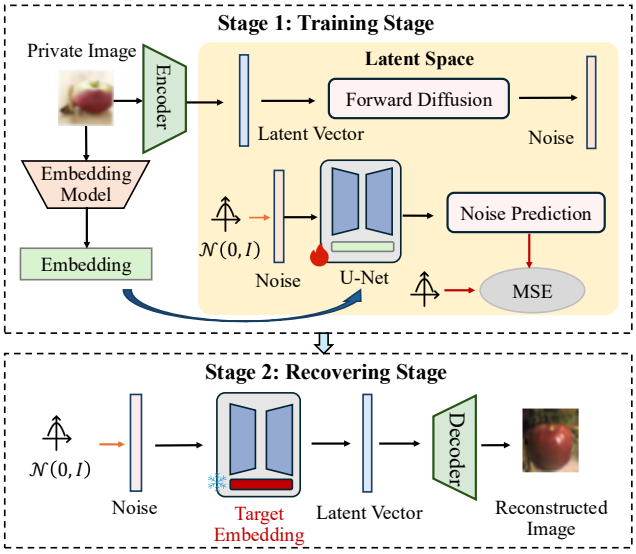


Figure 2: Overview of the proposed attack framework LIBRA.

embedding $e' = \mathbf{E}(x')$ using the black-box access to the embedding model $\mathbf{E}$. Subsequently, the adversary encodes x' into a latent vector $z'_0 \in \mathcal{Z}$ with the pretrained VQGAN encoder $\mathcal{E}$, i.e., $z'_0 = \mathcal{E}(x')$. Then, it trains a conditional diffusion model G_θ to learn $p_\theta(z'_0 \mid e')$. The sampling from this model can be denoted as $z \sim G_\theta(\cdot \mid e')$.

In the recovering stage, given a target embedding e stored in the vector database, the adversary reconstructs the corresponding input using the trained conditional diffusion model G_θ. The process starts by sampling a Gaussian noise latent vector $\hat{z}_T \sim \mathcal{N}(0, I)$, which is fed into G_θ along with the target embedding e. The model then iteratively denoises the latent variable, conditioned on e, to produce a clean latent representation $\hat{z}_0$. Finally, it maps $\hat{z}_0$ back to the image space using the VQGAN decoder, which generates the reconstructed image $\hat{x} = \mathcal{D}(\hat{z}_0)$.

4.2 Training Stage

We now detail the training stage in LIBRA. We adopt the conditional diffusion model (CDM) [13] as the generative backbone due to its capability to produce high-fidelity, finely detailed samples via a gradual denoising process. We leverage an auxiliary dataset $\mathbf{D}_{\text{aux}}$ to train the model, so that the domains are aligned, and the generated samples can effectively approximate the features of private images. Our training stage consists of two main steps: latent-space forward diffusion and conditional backward denoising.

4.2.1 Latent-space Forward Diffusion. For each image $x' \in \mathbf{D}_{\text{aux}}$, we first compute the corresponding embedding $e' = \mathbf{E}(x')$ via black-box queries. Note that we aim to efficiently initiate inversion from an embedding e by learning the conditional distribution $p_\theta(z_0 \mid e)$, which captures the latent representation corresponding to the underlying source image. We project each image x' into a latent space $\mathcal{Z}$ using a pretrained VQGAN encoder $\mathcal{E}$, yielding $z'_0 = \mathcal{E}(x')$.

It is worth noting that we use VQGAN to ensure that the diffusion model operates in a structured, semantically meaningful latent space rather than directly in the high-dimensional pixel space. The

VQGAN encoder compresses an image into a latent representation that preserves the essential semantic content while reducing dimensionality.

For instance, compared to processing 256×256 pixel values, operating on a compact latent vector (e.g., $\mathbb{R}^{512}$) greatly reduces the complexity of the diffusion process. During generation, the denoised latent can be mapped back to the image space by the VQGAN decoder to reconstruct high-fidelity outputs.

Next, we apply the standard DDPM forward noising process $\{z'_t\}_{t=0}^{T}$ in this latent space. Specifically, we specify a variance schedule $\{\beta_t\}_{t=1}^{T}$ with $\beta_t \in (0,1)$ and define $\alpha_t = 1 - \beta_t$ and the cumulative product $\bar{\alpha}_t = \prod_{i=1}^{t} \alpha_i$.

At each timestep t, we construct a Markov chain by adding Gaussian noise to the latent vector. The conditional distribution $q(z'_t \mid z'_{t-1})$ is defined as a Gaussian whose mean is a scaled version of the previous latent, $\sqrt{\alpha_t}\, z'_{t-1}$, and whose covariance is proportional to the identity matrix with variance β_t. Intuitively, this means that at every step we slightly attenuate the signal from z'_{t-1} and add a small amount of isotropic Gaussian noise. As t increases, repeated application of this noisy transition progressively destroys the information in the original latent z'_0, until z'_T becomes close to pure Gaussian noise, following the standard DDPM forward process.

4.2.2 Conditional Backward Denoising. The main objective of the backward prediction network is to estimate, at each timestep, the noise that was added during the forward diffusion steps and thereby realize a reverse denoising trajectory in the latent space. We use z'_t to denote latent from the auxiliary domain $\mathbf{D}_{\text{aux}}$ used for supervision, and z_t to denote the latent being reconstructed in the backward denoising step. Concretely, given a noisy latent vector z_t and a conditional embedding e', the model predicts a noise term $\epsilon_\theta(z_t, t, e')$, which can be interpreted as an estimate of the Gaussian perturbation that transformed into a cleaner latent z_{t-1} at timestep t.

This predicted noise is then used to define a Gaussian reverse transition distribution $p_\theta(z_{t-1} \mid z_t, e')$ whose mean is obtained by subtracting the estimated noise from z_t and rescaling it according to the same variance schedule $\{\alpha_t, \bar{\alpha}_t\}$ as in the forward process. The covariance Σ_t of this Gaussian distribution is chosen as a fixed diagonal matrix determined by the diffusion schedule.

We design the noise-prediction function ϵ_θ with a U-Net architecture for backward denoising. In contrast to LDM, where the condition is typically text or layout tokens and the goal is diverse synthesis, our condition (the image embedding) comes from an external target encoder and the goal is to reconstruct the original image from the embedding.

To achieve this, we present an embedding-guided cross-attention mechanism that uses the embedding as conditional signals to steer the generation process.

Embedding-guided Cross Attention Mechanism. We design a U-Net structure (as shown in Figure 3), which is composed of a sequence of cross-attention and convolutional blocks. On the encoder side, the three `CrossAttnDown` blocks take the noisy latent as *Queries* and the conditional embedding as *Keys* and *Values*. This enables the latent features to incorporate conditional guidance while reducing the spatial resolution. A subsequent `DownBlock` further downsamples these features and increases the channel capacity without additional conditioning. At the bottleneck, the `CrossAttnMid` block again performs cross-attention between the embedding and the latent representation to enrich the most compressed features. During decoding, the `UpBlock` restores the spatial resolution via upsampling and skip connections from the corresponding encoder stages. Finally, the three `CrossAttnUp` blocks inject the conditional signal once more at multiple scales through cross-attention, refining fine-grained details and completing the reconstruction path. Collectively, these cross-attention modules ensure that the conditional embedding is effectively integrated into the latent features throughout the U-Net.

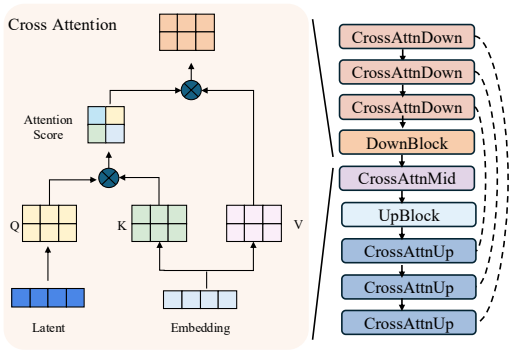


Figure 3: U-Net structure of the embedding-guided cross-attention mechanism.

Specifically, the formula for the cross-attention module is:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right) V. \tag{5}$$

With this U-Net architecture, we next outline the training objective within the VQGAN latent space that drives the denoising process. The initial latent input is sampled from a standard normal distribution $\mathcal{N}(0, I)$. A U-Net-based denoising model is trained to predict the added noise at each diffusion step. The model is optimized by minimizing the Mean Squared Error (MSE) between the predicted noise and the true Gaussian noise.

Typically, during training, we only need to compute the MSE loss between the predicted noise and the true noise $\|\epsilon - \epsilon_\theta(z_t, t)\|\, 2^2$ to train the model, where ϵ represents the Gaussian noise added from step 0 to step t, and $\epsilon_\theta(z_t, t)$ denotes the noise predicted by the generator G_θ based on z_t and t. However, in our work, we need to guide the generation process using our embedding as conditional information.

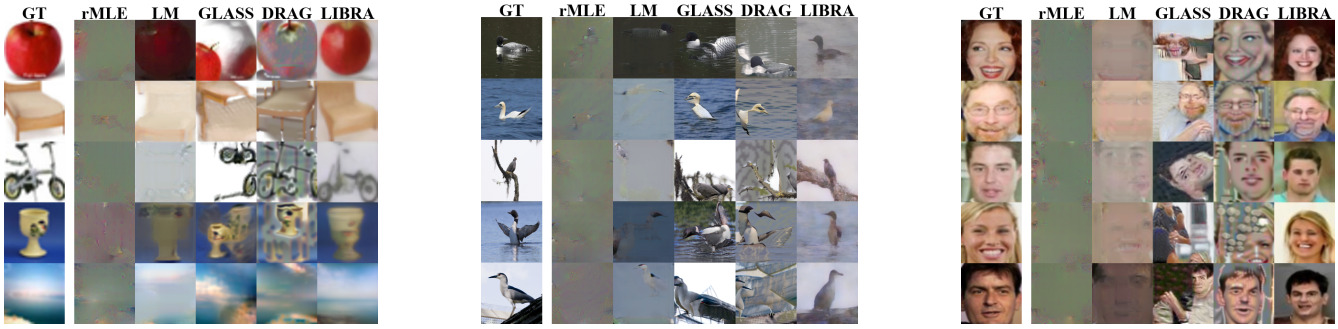
Therefore, during the training phase, the embedding e' is incorporated into ϵ_θ to train the model, then the MSE loss is computed between the true Gaussian noise and the noise predicted by the conditional diffusion model, which is $\left\|\epsilon - \epsilon_\theta(z'_t, t, e')\right\|_2^2$. This process is iteratively optimized with the following objective:

$$\min_\theta \mathcal{L}_{\text{MSE}}(\theta) = \mathbb{E}_{x' \sim \mathbf{D}_{\text{aux}},\, t,\, \epsilon \sim \mathcal{N}(0,I)} \left[\left\| \epsilon - \epsilon_\theta(z'_t, t, e') \right\|_2^2 \right], \tag{6}$$

The loss function in Eqn (6) shifts the training objective from the LDM goal of modeling $p(z \mid y)$ for diverse synthesis to instance-level inversion. It does not rely on pixel-space reconstruction or perceptual losses. It also avoids the use of gradients or intermediate activations, remaining fully compatible with a black-box setting.

Table 1: Comparison of reconstruction results on CIFAR-100 test set, NABirds, and LFW datasets.

Dataset	CIFAR-100 test set				NABirds				LFW			
Method	PSNR↑	SSIM↑	LPIPS↓	DINO↑	PSNR↑	SSIM↑	LPIPS↓	DINO↑	PSNR↑	SSIM↑	LPIPS↓	DINO↑
rMLE	10.23	0.63	0.89	0.28	11.23	0.57	0.77	0.17	12.60	0.61	0.94	0.18
LM	11.96	0.68	0.56	0.59	12.00	**0.61**	0.68	0.58	**14.96**	**0.65**	0.60	0.63
GLASS	11.01	0.60	0.61	0.60	11.53	0.44	**0.53**	**0.74**	10.57	0.45	0.65	0.38
DRAG	10.63	0.58	0.61	0.70	10.89	0.41	0.64	0.51	12.84	0.47	0.57	0.60
LIBRA	**14.89**	**0.70**	**0.45**	**0.79**	**12.72**	0.56	0.56	0.51	12.01	0.51	**0.51**	**0.81**



(a) CIFAR-100 test set (b) NABirds (c) LFW

Figure 4: Visual comparison across different methods on CIFAR-100 test set, NABirds and LFW using CLIP-ViT-B/32.

Algorithm 1: Training Phase in LIBRA

1 **Input**: noise schedule $\{\beta_t\}_{t=1}^{T}$, the number of steps N, batch size B, learning rate γ.
2 **Output**: Trained parameters θ^*.
1: Initialize parameters θ
2: Set $\alpha_t \leftarrow 1-\beta_t$ and $\bar{\alpha}_t \leftarrow \prod_{s=1}^{t} \alpha_s$ for all $t = 1, \ldots, T$
3: **for** iteration $n = 1, 2, \ldots, N$ **do**
4: **for** $j = 1, 2, \ldots, B$ **do**
5: $e'_j \leftarrow \mathbf{E}\left(x'_j\right)$
6: $z_0'^{(j)} \leftarrow \mathcal{E}\left(x'_j\right)$
7: Sample $t_j \sim \mathrm{Unif}\{1, \ldots, T\}$, $\xi_j \sim \mathcal{N}(0, I)$
8: $z_t'^{(j)} \leftarrow \sqrt{\bar{\alpha}_{t_j}}\, z_0'^{(j)} + \sqrt{1-\bar{\alpha}_{t_j}}\, \xi_j$
9: $\epsilon'_j \leftarrow \epsilon_\theta\left(z_t'^{(j)},\, t_j,\, e'_j\right)$
10: **end for**
11: Compute loss: $\mathcal{L} \leftarrow \frac{1}{B} \sum_{i=1}^{B} \left\| \epsilon_j - \epsilon'_j \right\|_2^2$
12: Update parameters: $\theta \leftarrow \theta - \gamma \cdot \nabla_\theta \mathcal{L}$
13: **end for**
14: **return** $\theta^* \leftarrow \theta$;

4.2.3 Training Algorithm Description. Algorithm 1 outlines the detailed training algorithm. During the forward diffusion stage (lines 7–8), each latent in the batch is assigned a randomly sampled diffusion timestep and perturbed with Gaussian noise according to the predefined variance schedule. This produces a noisy latent whose signal gradually diminishes as the assigned timestep increases. The reverse process is learned by training a denoising network to predict the exact noise injected in the forward diffusion stage. For each noisy latent, the model receives the perturbed latent, its corresponding timestep, and the associated embedding, and outputs an estimate of the added noise (line 9). The network parameters are then updated with gradient descent to minimize the mean squared error between the predicted and true noise across the batch (lines 11–12).

4.3 Recovering Stage

After training the conditional diffusion model, we can reconstruct the private image based on the embedding e. Instead of using the traditional diffusion sampling process for image generation, we perform conditional diffusion sampling directly in the latent space. At each timestep t, the model predicts the conditional noise $\epsilon_\theta(\hat{z_t}, t, e)$. The latent variable is updated from high noise to low noise following a predefined schedule, which yields stable, progressive denoising and reconstruction.

Specifically, given a target embedding e, we first initialize $\hat{z_T} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$, and then iterate over a time schedule $\{t_i\}_{i=1}^{K}$ from high to low and update:

$$\hat{z_{t-1}} = \frac{1}{\sqrt{\alpha_t}} \left(\hat{z_t} - \frac{\beta_t}{\sqrt{1-\bar{\alpha}_t}}\, \epsilon_\theta(\hat{z_t}, t, e) \right) + \sqrt{\beta_t}\, \xi, \tag{7}$$

where $\xi \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ is the vector of random Gaussian noise. After K iterations, we obtain $\hat{z}_0$, which can be decoded by VQGAN to produce the reconstructed image $\hat{x}$. We shall empirically show that $\hat{x}$ achieves high reconstruction quality compared to state-of-the-art baselines in Section 5.

5 Experimental Evaluation

In this section, we first introduce the experimental setup, then evaluate our attack performance by comparing it with state-of-the-art baselines, and finally discuss several defense strategies.

5.1 Experimental Setup

Datasets. We evaluate our attack across diverse domains, including object recognition, image classification, and face recognition. Specifically, we use *CIFAR-100 train set* [18], *CUB-200-2011* [41], and *CelebA* [23] as auxiliary datasets to train our models, and then apply the trained models to assess reconstruction performance on *CIFAR-100 test set* [18], *NABirds* [40], and *LFW* [14], respectively. The descriptions of these datasets are shown in Appendix A.1.

Models. We utilize three large-scale visual encoders as the embedding models, namely *CLIP-ViT-B/32* [30], *DINOv2-Base* [29], and *ResNet-101* [10]. The statistics of these models are summarized in Table 6 (Appendix A.2).

Baselines and Metrics. We compare our method with four representative optimization-based EIAs: *rMLE* [11], *LM* [34], *GLASS* [22] and *DRAG* [19]. These baselines are chosen for their relevance to feature-space inversion and data reconstruction in privacy-sensitive machine learning settings. The detailed descriptions of these baselines are provided in Appendix A.3.

To quantify attack performance by measuring the similarity between reconstructed and ground-truth images, we use four widely adopted metrics: *Peak Signal-to-Noise Ratio (PSNR)*, *Structural Similarity Index Measure (SSIM)* [43], *Learned Perceptual Image Patch Similarity (LPIPS)* [47] and *DINO Similarity* [4]. Specifically, PSNR and SSIM are employed to quantify pixel-level and structural fidelity, whereas LPIPS and DINO Similarity are utilized to capture perceptual and semantic consistency. The metric details are provided in Appendix A.4.

Latent Space Backbones. To compress images into the latent space, we use pretrained VQGAN models from the Latent Diffusion Models (LDM) framework. For facial data, we adopt *CompVis/ldm-celebahq-256* (trained on CelebA-HQ) to ensure identity consistency. For general data, we employ *CompVis/ldm-super-resolution-4x-openimages* (trained on OpenImages) to leverage its robust reconstruction capabilities and broad coverage.

Implementation. We implement the proposed attack method in Python and conduct experiments on machines equipped with two NVIDIA Tesla V100, two NVIDIA GeForce RTX 5090, and one NVIDIA H100 GPUs, running on Ubuntu 20.04. We use PyTorch to train the conditional diffusion model.

5.2 Attack Effectiveness Evaluation

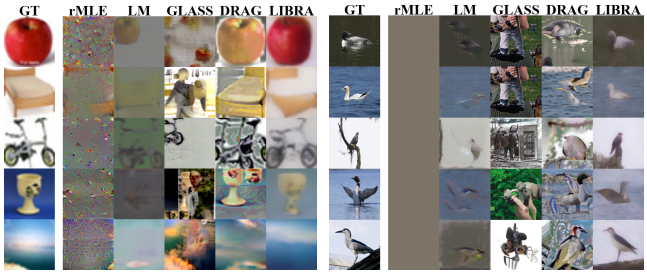
We first evaluate the reconstruction quality of our attack and compare it with four baselines, including rMLE, LM, GLASS, and DRAG.

Baseline Comparison. We compare the attack effectiveness against baselines using the CLIP-ViT-B/32 embedding model. We adopt a paired auxiliary–test configuration: the CIFAR-100 train set serves as the auxiliary source for the CIFAR-100 test set, CUB-200-2011 provides auxiliary supervision for NABirds, and CelebA is utilized for LFW. Table 1 reports the resulting metrics, and Figure 4 illustrates the visual reconstructions.

Overall, our attack method achieves favorable reconstruction quality across datasets. It attains the highest PSNR on CIFAR-100 test set and NABirds, while remaining competitive on the LFW dataset. Also, LIBRA yields the lowest LPIPS on CIFAR-100 test set and LFW. From the visualization results in Figure 4, we observe that

Table 2: Comparison of reconstruction results on CIFAR-100 test set with DINOv2-Base and NABirds with ResNet-101.

Dataset	CIFAR-100 test set			NABirds		
Method	PSNR↑	SSIM↑	LPIPS↓	PSNR↑	SSIM↑	LPIPS↓
rMLE	9.67	0.38	0.95	10.80	0.58	0.86
LM	10.02	0.65	0.71	13.34	0.61	0.59
GLASS	9.82	0.43	0.81	8.89	0.26	0.71
DRAG	10.35	0.51	0.62	10.94	0.36	0.64
LIBRA	**14.31**	**0.74**	**0.37**	**15.23**	**0.64**	**0.47**



(a) CIFAR-100 test set **(b) NABirds**

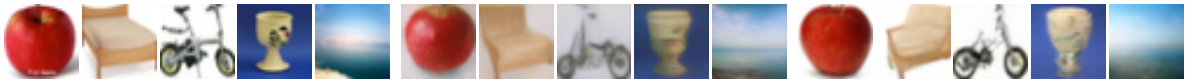
Figure 5: Visual comparison on CIFAR-100 test set with DINOv2-Base and NABirds with ResNet-101.

LIBRA produces visually coherent reconstructions that preserve more recognizable structures and semantic content. In particular, optimization-based approaches like rMLE and LM tend to produce over-smoothed, "shadow-like" images. While these methods achieve relatively high PSNR and SSIM by preserving coarse luminance and structure, they fail to recover valid textures, rendering the outputs perceptually deficient. In contrast, generative baselines such as GLASS and DRAG recover partial information. GLASS, constrained by domain-specific GAN priors, frequently suffers from geometric drift (e.g., head-torso disproportion in NABirds), whereas DRAG's weak conditioning on target embeddings leads to structural discontinuities and semantic inconsistencies.

These results indicate that our reconstructions not only resemble the private images in pixel-level fidelity but also maintain significantly stronger semantic alignment.

Embedding Model Generalization. To systematically evaluate the cross-domain generalization of our attack, we further conduct experiments on the other two embedding models: self-supervised DINOv2-Base and the convolutional ResNet-101. The results are shown in Table 2 and Figure 5, demonstrating the robustness of our attack. Specifically, on the CIFAR-100 dataset using a DINOv2-Base encoder, LIBRA outperforms the strongest baseline with a 38.2% improvement in PSNR and a 40.3% reduction in LPIPS. Similarly, on the NABirds dataset with a ResNet-101 backbone, LIBRA achieves a 14.2% gain in PSNR. These findings suggest that our attack can effectively adapt to diverse embedding distributions without necessitating architecture-specific modifications.

Our experiments further reveal a distinct hierarchy of reconstruction difficulty across different embedding models. CNN-based architectures (e.g., ResNet-101) exhibit the highest vulnerability, as their local receptive fields and hierarchical feature maps preserve



(a) GT (b) Latent-space Reconstruction (c) Pixel-space Reconstruction

Figure 6: Visual comparison of reconstruction quality across different reconstruction strategies on CIFAR-100.

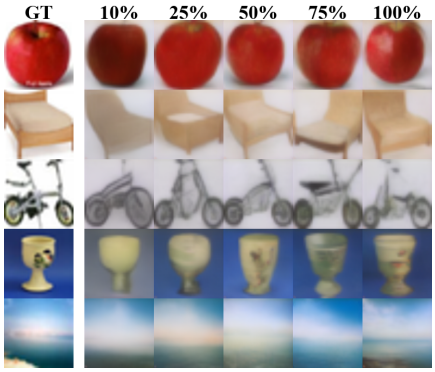


Figure 7: Impact of auxiliary data size on attack quality.

Table 3: Per-target recovery time comparison for different embedding models. For LIBRA, the reported time denotes the recovery time after the one-time query and training stages.

Method	# of Iterations	CLIP-ViT-B/32	DINOv2-Base	ResNet-101
rMLE	20,000	24 min 59 s	30 min 54 s	31 min 03 s
LM	20,000	27 min 28 s	44 min 59 s	41 min 35 s
GLASS	20,000	1 hr 31 min 58 s	1 hr 41 min 29 s	1 hr 12 min 00 s
DRAG	4,000	1 hr 37 min 33 s	1 hr 37 min 33 s	1 hr 14 min 12 s
LIBRA	**1,000**	**3 min 47 s**	**2 min 10 s**	**3 min 47 s**

substantial low-level structural cues. Conversely, vision transformers utilizing global attention (e.g., DINOv2 and CLIP) yield more abstract representations that are marginally more resilient to structural recovery.

Commercial Multimodal Model. We further evaluate LIBRA on Alibaba Cloud Bailian, a commercial multimodal embedding model, using CIFAR-100 as the target dataset. The attack remains effective, achieving PSNR of 13.67, SSIM of 0.51, LPIPS of 0.48, and DINO similarity of 0.65. This result broadens our evaluation beyond open-source encoders and shows that LIBRA can also pose privacy risks to practical cloud-based multimodal embedding services.

5.3 Attack Efficiency Evaluation

Next, we evaluate the efficiency of initiating the attack and compare it with baselines. As summarized in Table 3, LIBRA achieves substantially faster per-image recovery than existing baselines. Specifically, once the conditional diffusion model is trained, LIBRA reduces the wall-clock recovery time by 85%–95% compared to optimization-based approaches such as rMLE and LM, and by over 95% compared to GAN- or pixel-space diffusion attacks such as GLASS and DRAG.

To provide a more complete view of the attacker's cost, we further decompose the end-to-end computation time as

$$T_{\text{total}}^{(\mathcal{M})} = T_{\text{query}}^{(\mathcal{M})} + T_{\text{train}}^{(\mathcal{M})} + N_{\text{target}} \cdot T_{\text{recover}}^{(\mathcal{M})}, \tag{8}$$

where $T_{\text{query}}^{(\mathcal{M})}$ and $T_{\text{train}}^{(\mathcal{M})}$ denote the one-time query and training costs under embedding model $\mathcal{M}$, and $T_{\text{recover}}^{(\mathcal{M})}$ denotes the per-target recovery time. For example, using CLIP-ViT-B/32, the total cost of LIBRA can be expressed as

$$T_{\text{total}}^{(\text{CLIP})} = 403517.5 + 227 \cdot N_{\text{target}}, \tag{9}$$

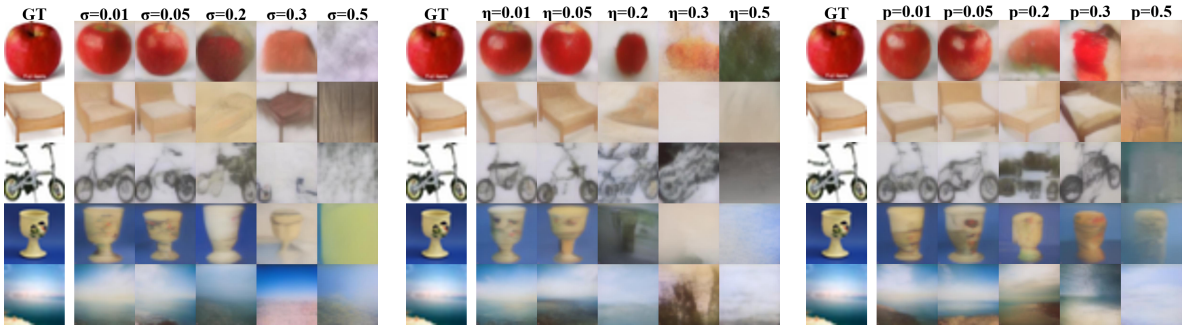
where the time is measured in seconds. Here, 403,517.5 seconds corresponds to the one-time query and training cost, and 227 seconds corresponds to the per-target recovery time reported in Table 3. This decomposition shows that the one-time query and training cost dominates the total cost when the number of target images is small. However, after the model has been trained, the marginal cost of reconstructing each additional image is low. Therefore, LIBRA is especially efficient when recovering a large number of target images, as it avoids the expensive per-sample optimization required by prior methods. This efficiency mainly comes from performing conditional diffusion in the compact VQGAN latent space and using the embedding-guided U-Net to directly steer the denoising trajectory.

5.4 Ablation Study

Impact of VQGAN Encoder. We investigate the efficacy of the VQGAN latent space through an ablation study on CIFAR-100 (CLIP-ViT-B/32), benchmarking against a pixel-space baseline. As shown in Table 5 and Figure 6, both reconstruction strategies yield comparable reconstruction fidelity, with latent-space inversion achieving a slightly higher PSNR and SSIM. While the pixel-space variant exhibits marginal gains in perceptual alignment (e.g., 8.9% lower LPIPS and 2.5% higher DINO similarity), latent-space reconstructions preserve clearer object contours and more stable structural layouts. Crucially, by operating within the compressed VQGAN manifold, the latent-space approach reduces training wall-clock time by 63.7%, offering a better trade-off between reconstruction quality and computational efficiency.

Impact of Embedding-guided Cross Attention. We further investigate the effectiveness of the proposed embedding-guided cross-attention mechanism. To this end, we replace the cross-attention module with a direct embedding concatenation strategy, where the target embedding is simply concatenated with the latent feature representation as the conditional input. As shown in Table 4, removing the embedding-guided cross-attention mechanism leads to a substantial degradation across all metrics. Specifically, PSNR drops from 14.89 to 7.72, SSIM decreases from 0.70 to 0.42, LPIPS increases from 0.45 to 0.79, and DINO similarity decreases from 0.79 to 0.38. These results demonstrate that naive embedding concatenation is insufficient for effectively injecting the target embedding into the diffusion process. In contrast, our embedding-guided cross-attention mechanism enables the model to align the denoising trajectory with the semantic and structural information encoded in the target embedding, thereby producing more faithful reconstructions.

Impact of Auxiliary Data Size. We evaluate the reconstruction quality by varying the auxiliary training data from 10% to 100%. As illustrated in Figure 7, reconstruction fidelity remains robust even in low-data regimes. These observations are substantiated by



(a) Gaussian Noise Injection (b) Laplacian Noise Injection (c) Dimensional Masking

Figure 8: Qualitative comparison of embedding-level perturbation defenses.

Table 4: Ablation study on the embedding-guided cross-attention mechanism on CIFAR-100 using CLIP-ViT-B/32.

Setting	PSNR↑	SSIM↑	LPIPS↓	DINO↑
w/ cross-attention	14.89	0.70	0.45	0.79
w/o cross-attention	7.72	0.42	0.79	0.38

Table 5: Quantitative comparison between pixel-space and latent-space inversion on CIFAR-100.

Setting	PSNR↑	SSIM↑	LPIPS↓	DINO↑	Train Time ↓
latent-space	**14.89**	**0.70**	0.45	0.79	**8 min 13 s**
pixel-space	14.43	0.68	**0.41**	**0.81**	22 min 38 s

the quantitative results in Figure 9 (Appendix C), which show a consistent upward trend where PSNR improves from 13.08 to 14.89 and DINO rises from 0.63 to 0.79 as data scales from 10% to 100%. Besides, LIBRA exhibits strong data efficiency: using only 25% of the data achieves a PSNR of 14.04, recovering most of the full-data performance. The metric saturation observed beyond 50% suggests the model's ability to learn effective domain priors with limited supervision.

5.5 Defense Strategies

Finally, we examine three defense mechanisms to analyze the trade-off between privacy protection and reconstruction quality. Figure 8 and Table 8 (Appendix D) summarize the effect of three perturbation schemes on CIFAR-100 test set (using CLIP-ViT-B/32): Gaussian noise with standard deviation σ, Laplacian noise with scale η, and dimensional masking with dropout rate p. A general degradation trend is observed across the three strategies as the perturbation intensity increases.

- **Weak Perturbation ($\sigma, \eta, p \leq 0.10$):** Gaussian and Laplacian noise largely preserve reconstruction quality under small perturbation strengths, while dimensional masking causes a more noticeable degradation in semantic similarity even at low masking rates.
- **Moderate Perturbation ($\sigma, \eta, p \approx 0.20$–$0.30$):** Reconstruction quality degrades more clearly. DINO similarity decreases noticeably, and LPIPS increases, indicating weaker perceptual and semantic alignment. Nevertheless, coarse object geometry and dominant color distributions often remain distinguishable.
- **Strong Perturbation ($\sigma, \eta, p = 0.50$):** All three defenses substantially degrade reconstruction quality. DINO similarity drops by roughly 37%–42%, and LPIPS increases by approximately 44%–62% compared with the undefended baseline, making the recovered images much less informative.

Quantitatively, as perturbation strength escalates, PSNR and DINO similarity decrease while LPIPS increases. Notably, when perturbation exceeds 0.30, fine textures and contours are obliterated, leaving only coarse silhouettes. This confirms that all three defenses effectively mitigate privacy risks by degrading the perceptual fidelity of recovered images. Regarding artifact types, Gaussian noise primarily induces global blurring, Laplacian noise generates grainy artifacts and localized distortions, and dimensional masking leads to feature collapse and semantic loss, collectively presenting diverse challenges to the robustness of reconstruction models.

6 Conclusion

In this paper, we propose a novel and practical black-box image embedding inversion attack LIBRA, which reconstructs private images directly from outsourced embeddings. We present an embedding-guided cross-attention mechanism to train a conditional diffusion model that effectively aligns diffusion trajectories with the target embedding. To improve efficiency, we perform diffusion in the compact latent space of a pretrained VQGAN, enabling fast training and inference while preserving structural and perceptual details. We conduct extensive experiments on three real-world datasets and three embedding models to evaluate LIBRA. The results demonstrate that LIBRA consistently surpasses four state-of-the-art inversion baselines. Our findings underscore the urgent need for privacy protection of the embeddings in outsourced vector databases.

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A Experimental Setup

In this section, we provide detailed specifications of the datasets, embedding models, baseline attacks, and evaluation metrics used in our study.

A.1 Datasets

We evaluate our method across diverse visual domains using the following standard benchmarks:

- *CIFAR-100* [18] contains 60,000 natural images at 32×32 resolution across 100 classes (50,000 train / 10,000test).
- *CUB-200-2011* [41] contains 11,788 bird images from 200 species with fine-grained annotations (parts and attributes).
- *CelebA* [23] contains 202,599 face images of 10,177 identities with 40 annotated attributes and 5 landmarks.
- *NABirds* [40] contains 48,562 images of 555 North American bird species, providing fine-grained annotations for bird species recognition.
- *LFW* [14] contains 13,233 "in-the-wild" face images of 5,749 identities, collected under unconstrained conditions (e.g., varying pose, illumination, and background).

A.2 Models

We target three distinct visual encoders to evaluate cross-architecture generalization. Their statistics are summarized in Table 6.

- *CLIP-ViT-B/32*[1]: a Vision Transformer model trained with contrastive language–image pretraining. We use its image encoder to obtain 512-D normalized embeddings.
- *DINOv2-Base*[2]: a ViT-Base encoder trained with DINOv2 self-distillation on large curated data. We use its image encoder to obtain 768-D normalized embeddings.
- *ResNet-101* [10]: a 101-layer residual CNN trained on ImageNet with supervised labels. We use the global-average-pooled 2048-D features as embeddings.

A.3 Baseline Attacks

We compare our approach against four state-of-the-art embedding inversion attacks:

- *rMLE* [11]. The first optimization-based attack that searches for an image whose client-side feature matches the target representation, while a total-variation prior encourages natural-looking results.
- *LM* [34]. Builds on rMLE with a deep image prior by optimizing a CNN generator fed with fixed noise so that its output matches the target feature. The total-variation prior is retained to suppress artifacts.
- *GLASS* [22]. A GAN-based attack that performs a coarse search in StyleGAN latent z-space with KL and TV regularizers, then refines in the $W+$ space for detail, where each generator layer is assigned an independent style vector. It uses domain-specific priors: StyleGAN2-ADA[3] for faces and StyleGAN-XL[4] for other categories, which brings an advantage when the data domain is known.
- *DRAG* [19]. A guided-diffusion attack that reconstructs images from deep intermediate features by reformulating feature matching as classifier-style guidance. The method incorporates Tweedie updates, a spherical-Gaussian constraint, gradient clipping, and a self-loop to improve stability. It adopts Stable Diffusion v1.5[5] as a single, domain-agnostic prior.

Table 6: Statistics of the embedding models used in the experiments.

Model	Params	Layers	Year	Developer	Type
CLIP-ViT-B/32	88.2M	12	2021	OpenAI	Vision Transformer(Multimodal Encoder)
DINOv2-Base	86M	12	2023	Meta AI	Self-supervised Vision Transformer
ResNet-101	44.5M	101	2015	MSR	CNN(Image Encoder)

A.4 Metrics

We quantify reconstruction quality using four standard metrics:

- *Peak Signal-to-Noise Ratio (PSNR)*, which assesses pixel-level fidelity between reconstructed and ground-truth images, where a higher value indicates more accurate reconstruction.
- *Structural Similarity Index Measure (SSIM)* [43], which measures perceived structural consistency and luminance similarity between two images. Higher SSIM values reflect better preservation of visual structure.
- *Learned Perceptual Image Patch Similarity (LPIPS)* [47], which quantifies perceptual similarity in a deep feature space extracted by pretrained networks, with lower scores corresponding to greater perceptual closeness.
- *DINO Similarity* [4], which captures semantic alignment in a self-distilled representation space, where higher similarity denotes stronger correspondence in high-level visual semantics.

A.5 Implementation Details

We train a conditional U-Net with multi-scale cross-attention in the latent space of a pretrained VQGAN. Input images are resized to 64×64 and normalized to $[-1, 1]$, then encoded into latents. Diffusion is performed over T=1000 DDPM steps with a scaled-linear β schedule and ϵ-prediction. We optimize the model with AdamW (learning rate 1×10^{-4}) and use a cosine learning-rate schedule. During evaluation, we initialize reconstruction from a Gaussian noise latent. The latent is iteratively denoised by the reverse diffusion process, conditioned on the target embedding at each step, and is finally decoded to the RGB space using the VQGAN decoder.

[1] https://huggingface.co/openai/clip-vit-base-patch32
[2] https://huggingface.co/facebook/dinov2-base
[3] https://github.com/NVlabs/stylegan2-ada-pytorch
[4] https://github.com/autonomousvision/stylegan-xl
[5] https://huggingface.co/stable-diffusion-v1-5/stable-diffusion-v1-5

Table 7: Reconstruction performance under in-distribution and out-of-distribution auxiliary data settings on CIFAR-100 using CLIP-ViT-B/32.

Setting	PSNR↑	SSIM↑	LPIPS↓	DINO↑
In-distribution	14.89	0.70	0.45	0.79
OOD	12.36	0.55	0.61	0.49

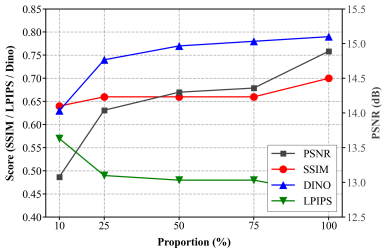


Figure 9: Impact of auxiliary dataset size on attack quality.

B Impact of Out-of-Distribution Auxiliary Data

In our main experiments, we assume that the auxiliary dataset is drawn from the same or a similar distribution as the private images. To further evaluate the robustness of LIBRA under a more challenging threat model, we conduct an additional experiment using out-of-distribution (OOD) auxiliary data. Specifically, we use ImageNet-1K as a large-scale OOD auxiliary dataset and CLIP-ViT-B/32 as the target embedding model to reconstruct CIFAR-100 target images. We randomly sample 100 images from the CIFAR-100 test set as recovery targets and report the average reconstruction performance.

As shown in Table 7, using OOD auxiliary data leads to a moderate degradation in reconstruction quality across all four metrics. This drop is expected, since ImageNet-1K does not fully match the distribution of CIFAR-100 target images, making it harder for the conditional diffusion model to learn domain-aligned priors. Nevertheless, the recovered samples still preserve object contours, global layout, and dominant color patterns, and remain semantically recognizable from the visual perspective. These results indicate that LIBRA can still recover meaningful information even when the auxiliary dataset is out-of-distribution, suggesting that the privacy risk persists under distribution mismatch.

C Impact of Auxiliary Data Size

We quantitatively analyze the impact of auxiliary dataset size on attack performance in Figure 9. As the data proportion scales from 10% to 100%, we observe a consistent improvement across all metrics: PSNR (black curve) rises from 13.08 dB to 14.89 dB, and the semantic consistency metric DINO (blue curve) increases from 0.63 to 0.79. Crucially, the performance exhibits a steep ascent in the low-data regime; using only 25% of the data, the model already recovers the majority of its performance (14.04 dB PSNR and 0.74 DINO).

The subsequent saturation trends, particularly visible in SSIM (red) and LPIPS (green) beyond the 25%-50% range, underscore the high data efficiency of our proposed method.

Table 8: Quantitative comparison of embedding-level perturbation defenses.

(a) Gaussian Noise Injection

σ	PSNR↑	SSIM↑	LPIPS↓	DINO↑
0.01	14.48	0.67	0.48	0.78
0.05	14.08	0.68	0.46	0.78
0.10	14.28	0.67	0.47	0.75
0.20	14.18	0.70	0.50	0.70
0.30	13.11	0.69	0.54	0.58
0.50	12.03	0.64	0.67	0.50

(b) Laplacian Noise Injection

η	PSNR↑	SSIM↑	LPIPS↓	DINO↑
0.01	13.93	0.64	0.49	0.76
0.05	13.92	0.69	0.49	0.77
0.10	13.43	0.66	0.50	0.72
0.20	13.51	0.68	0.53	0.62
0.30	12.77	0.70	0.60	0.55
0.50	11.12	0.61	0.73	0.46

(c) Dimensional Masking

p	PSNR↑	SSIM↑	LPIPS↓	DINO↑
0.01	13.98	0.63	0.51	0.67
0.05	14.22	0.64	0.53	0.64
0.10	13.98	0.63	0.53	0.63
0.20	13.68	0.64	0.56	0.59
0.30	13.13	0.62	0.59	0.54
0.50	12.24	0.60	0.65	0.47

D Defense Strategies

We evaluate the robustness of our attack against three embedding-level defense mechanisms:

- **Gaussian noise injection** [1]: Injects i.i.d. zero-mean Gaussian noise $\mathcal{N}(0, \sigma^2)$ into each embedding dimension.
- **Laplacian noise injection** [8]: Injects i.i.d. zero-mean Laplace noise $\mathrm{Lap}(0, \eta)$ into each embedding dimension. The heavy-tailed distribution can introduce occasional larger deviations than Gaussian noise, and the scale parameter η controls the perturbation magnitude.
- **Dimensional masking** [37]: Randomly masks (sets to zero) a proportion of embedding dimensions to reduce exposed information and limit feature-reconstruction attacks.

Table 8 presents a detailed quantitative analysis of reconstruction quality under these perturbations. Specifically, for Gaussian and Laplacian noise injections, the DINO scores remain above 0.70 even when the noise intensity reaches 0.10, indicating that semantic information is well-preserved. For dimensional masking, PSNR remains relatively high up to a masking ratio of $p = 0.20$ with a value of 13.68 dB, but perceptual and semantic metrics already show noticeable degradation, as LPIPS increases to 0.56 and DINO similarity decreases to 0.59. The results indicate that our inversion method retains partial reconstruction ability under weak perturbations, but reconstruction quality degrades as the perturbation strength increases.